



SEMI CONDUCTORS PHYSICS

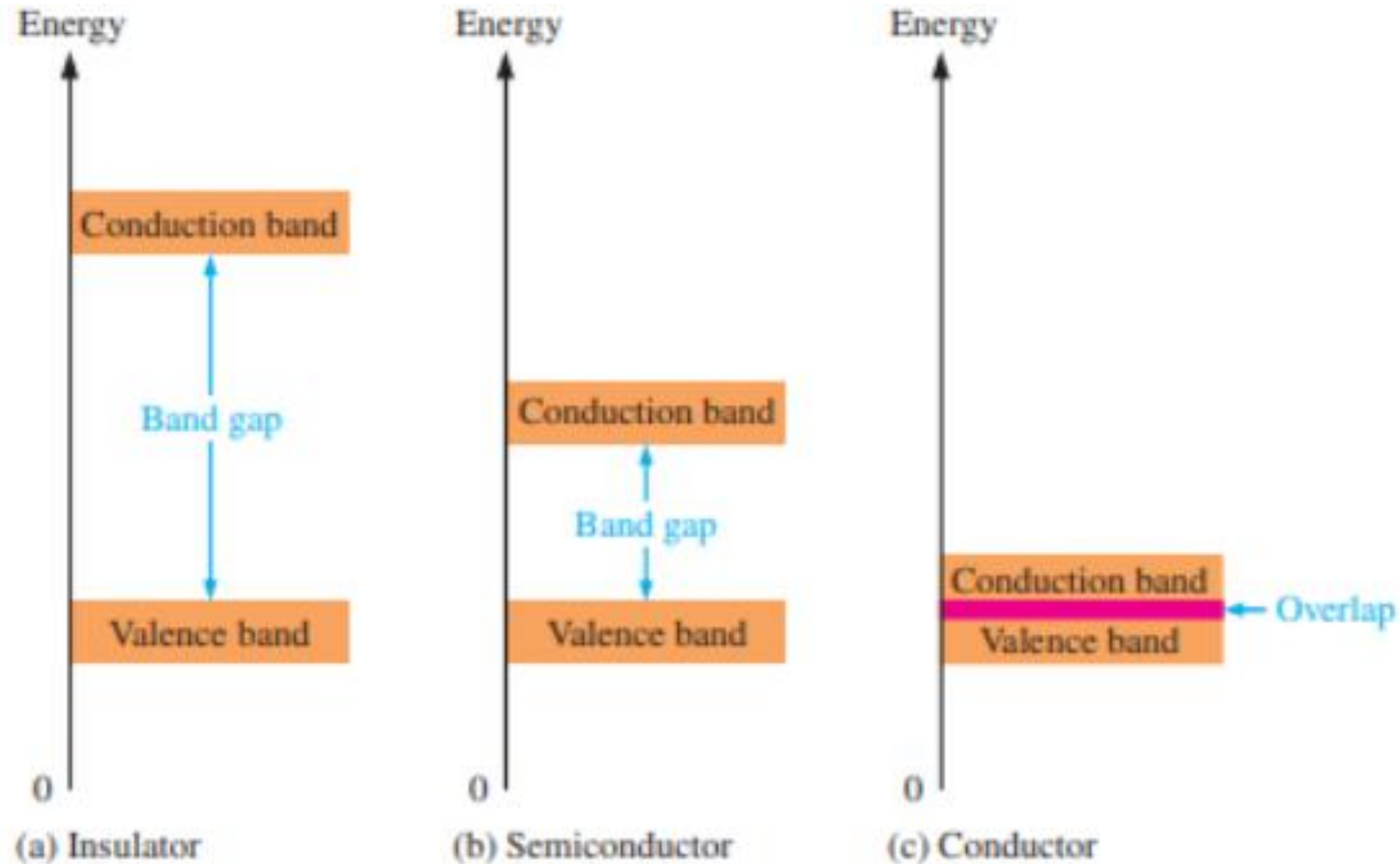


ENERGY BAND THEORY OF SOLIDS:

- ▶ The energy band which is formed by grouping the range of energy levels of the free electrons is called as conduction band.
- ▶ The energy gap which is present between the valence band and conduction band by separating these two energy bands is called as forbidden band or forbidden gap.
- ▶ The energy band which is formed by grouping the range of energy levels of the valence electrons or outermost orbit electrons is called as valence band.

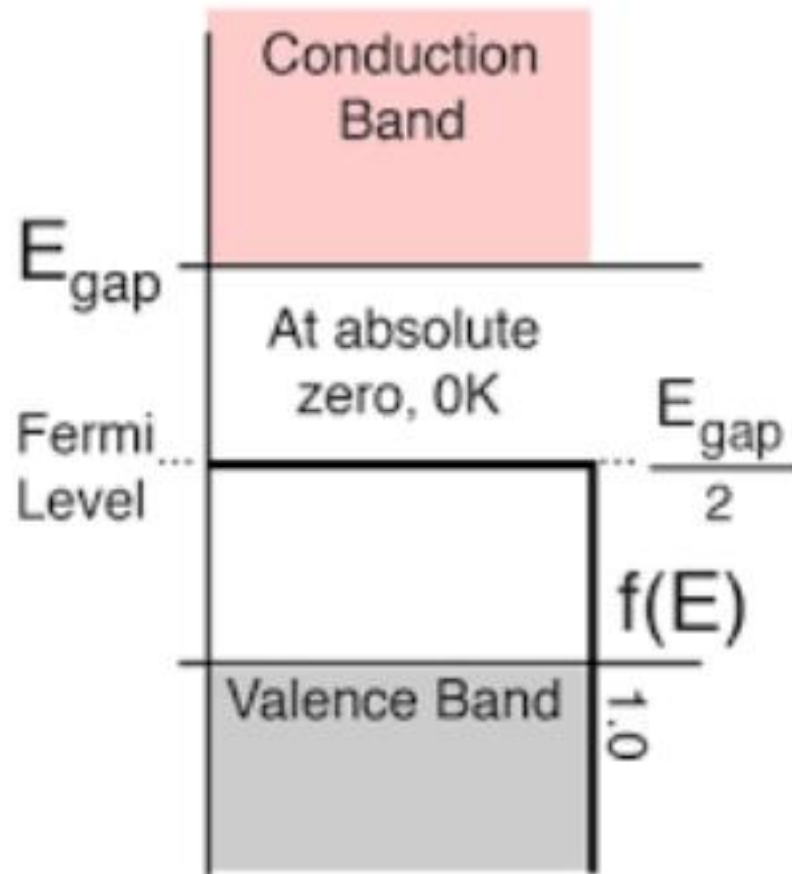


ENERGY BAND THEORY OF SOLIDS:



Fermi Energy:

The highest energy level that a electron can occupy at zero kelvin.



i. Insulators:

- ▶ In insulators the valence band is fully occupied with electrons due to sharing of outer most orbit electrons with the neighboring atoms.
- ▶ Whereas conduction band is empty, i.e., no electrons are present in conduction band.
- ▶ The forbidden gap between the valence band and conduction band is very large in insulators
- ▶ The electrons in valence band cannot move because they are locked up between the atoms. In order move the valence band electrons into conduction band large amount of external energy is applied which is equal to the forbidden gap. But in insulators, this is practically impossible to move the valence band electrons into conduction band.



ii. Semi Conductors:

- ▶ The material which has electrical conductivity between that of a conductor and an insulator is called as semiconductor.
 - ▶ In semiconductors, the forbidden gap between valence band and conduction band is very small.
 - ▶ At low temperature, the valence band is completely occupied with electrons and conduction band is empty because the electrons in the valence band does not have enough energy to move into conduction band. Therefore, semiconductor behaves as an insulator at low temperature.
 - ▶ However, at room temperature some of the electrons in valence band gains enough energy in the form of heat and moves into conduction band.
 - ▶ When the temperature increases, the number of valence band electrons moving into conduction band also increases. This shows that electrical conductivity of the semiconductor increases with increase in temperature
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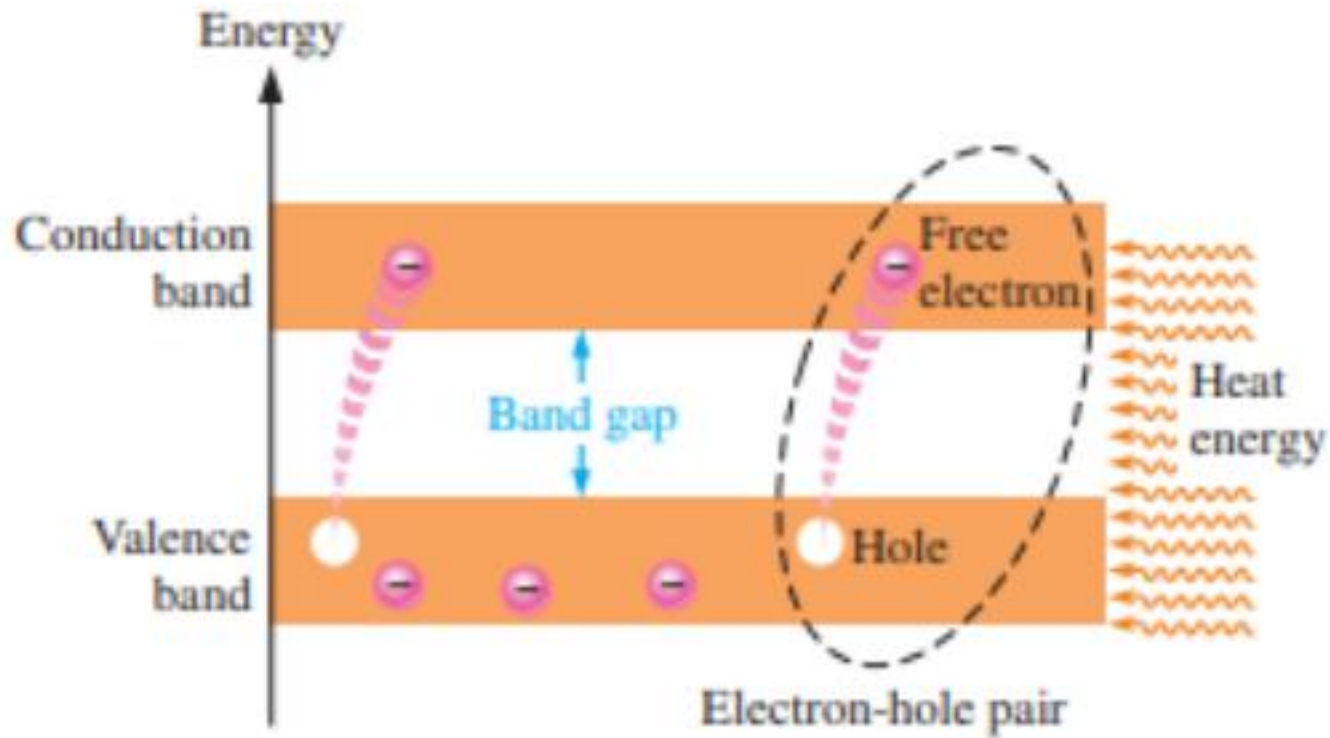


iii. Conductors:

- ▶ In a conductor, valence band and conduction band overlap each other as shown in figure. Therefore, there is no forbidden gap in a conductor.
 - ▶ A small amount of applied external energy provides enough energy for the valence band electrons to move into conduction band. Therefore, greater number of valence band electrons can easily move into the conduction band.
 - ▶ When valence band electrons moves to conduction band, they becomes free electrons. The electrons present in the conduction band are not attached to the nucleus of an atom.
 - ▶ In conductors, large number of electrons are present in conduction band at room temperature, i.e, conduction band is almost full of electrons. Whereas valence band is partially occupied with electrons. The electrons present in the conduction band moves freely by carrying the electric current from one point to other
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ELECTRON – HOLE CONCEPT:



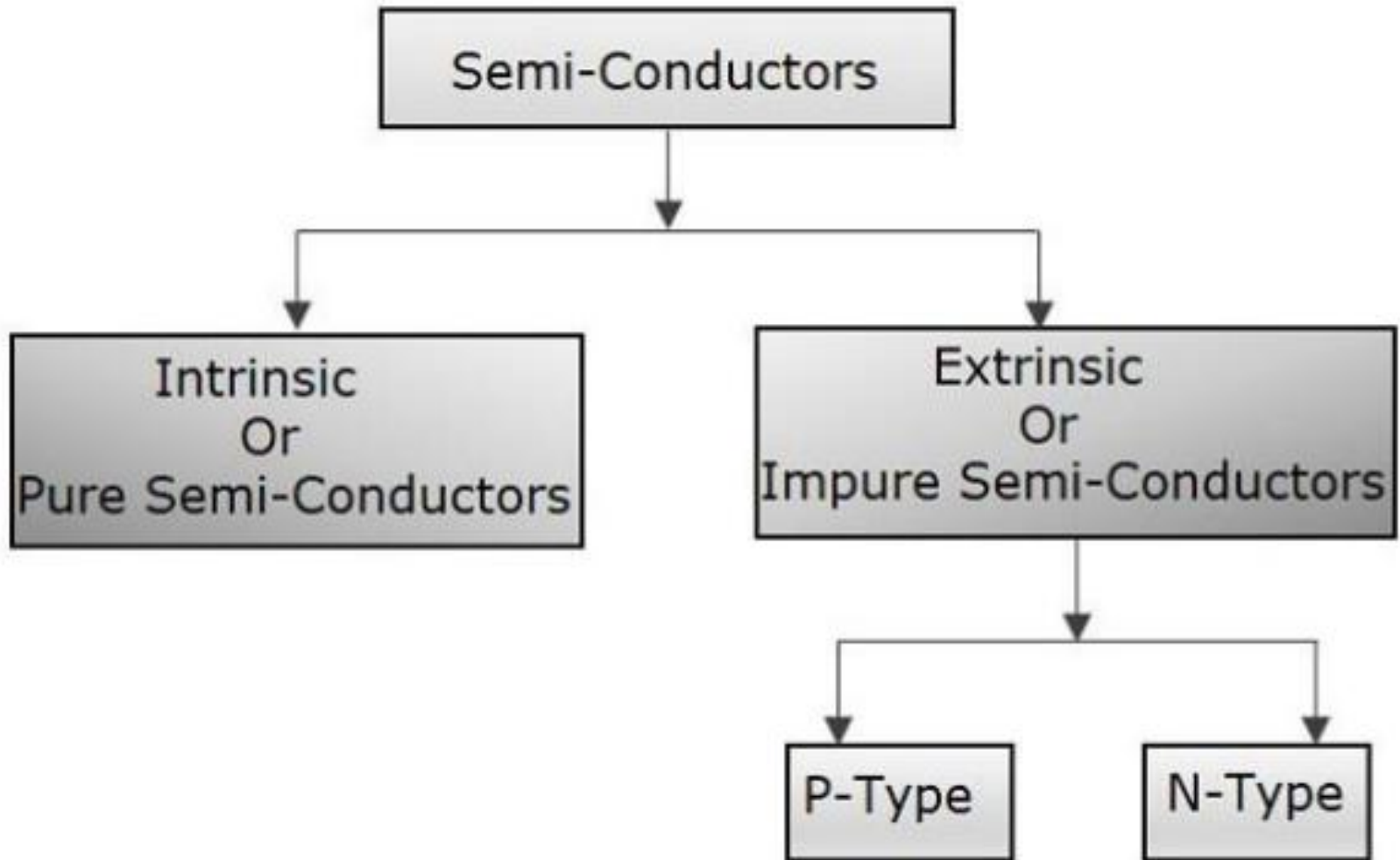
ELECTRON HOLE CONCEPT:

- ▶ An intrinsic (pure) silicon crystal at room temperature has sufficient heat (thermal) energy for some valence electrons to jump the gap from the valence band into the conduction band, becoming free electrons. Free electrons are also called **conduction electrons**.
- ▶ When an electron jumps to the conduction band, a vacancy is left in the valence band within the crystal. This vacancy is called a **hole**.
- ▶ For every electron raised to the conduction band by external energy, there is one hole left in the valence band, creating what is called an **electron-hole pair**.
- ▶ **Recombination** occurs when a conduction-band electron loses energy and falls back into a hole in the valence band.



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- ▶ When a voltage is applied across a piece of intrinsic silicon, the thermally generated free electrons in the conduction band (which are free to move randomly in the crystal structure) are attracted toward the positive end. This movement of free electrons in a semiconductive material is called **electron current**.
 - ▶ A valence electron can move into a nearby hole with little change in its energy level, thus leaving another hole where it came from. Effectively the hole has moved from one place to another in the crystal structure. Although current in the valence band is produced by valence electrons, it is called **hole current** to distinguish it from electron current in the conduction band.
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INTRINSIC SEMI CONDUCTORS:

- ▶ Semi conductive materials do not conduct current well because of the limited number of free electrons in the conduction band and holes in the valence band.
- ▶ Intrinsic silicon (or germanium) must be modified by increasing the number of free electrons or holes to increase its conductivity and make it useful in electronic devices. This is done by adding impurities to the intrinsic material.



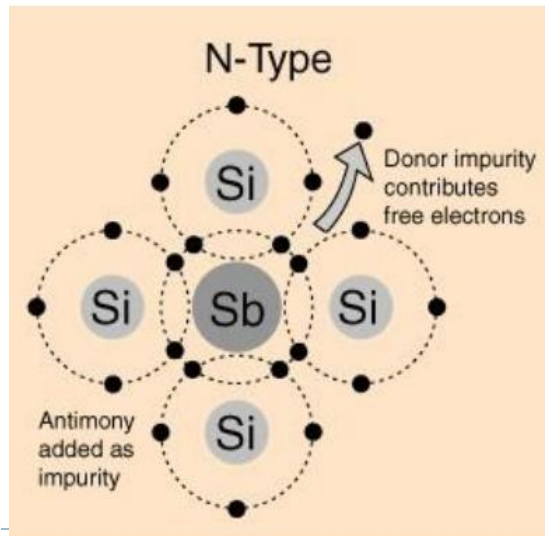
EXTRINSIC SEMI CONDUCTORS

- ▶ Two types of extrinsic (impure) semiconductive materials, n-type and p-type, are the key building blocks for most types of electronic devices.
- ▶ DOPING:
Semiconductors conductivity can be drastically increased by the controlled addition of impurities to the intrinsic (pure) semiconductive material, called doping,
- ▶ Doping increases the number of current carriers (electrons or holes).
- ▶ Types :
 - ▶ n-type (Pentavalent doping)
 - ▶ p-type (Trivalent doping).



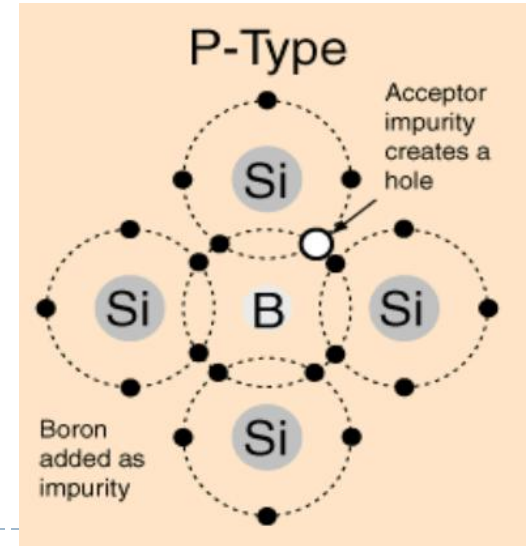
PENTAVALENT DOPING

- ▶ Doping occurs when Si or Ge is doped with an element of group V,
- ▶ Produces N-type semiconductors,
- ▶ Also called as donor doping.



TRIVALENT DOPING

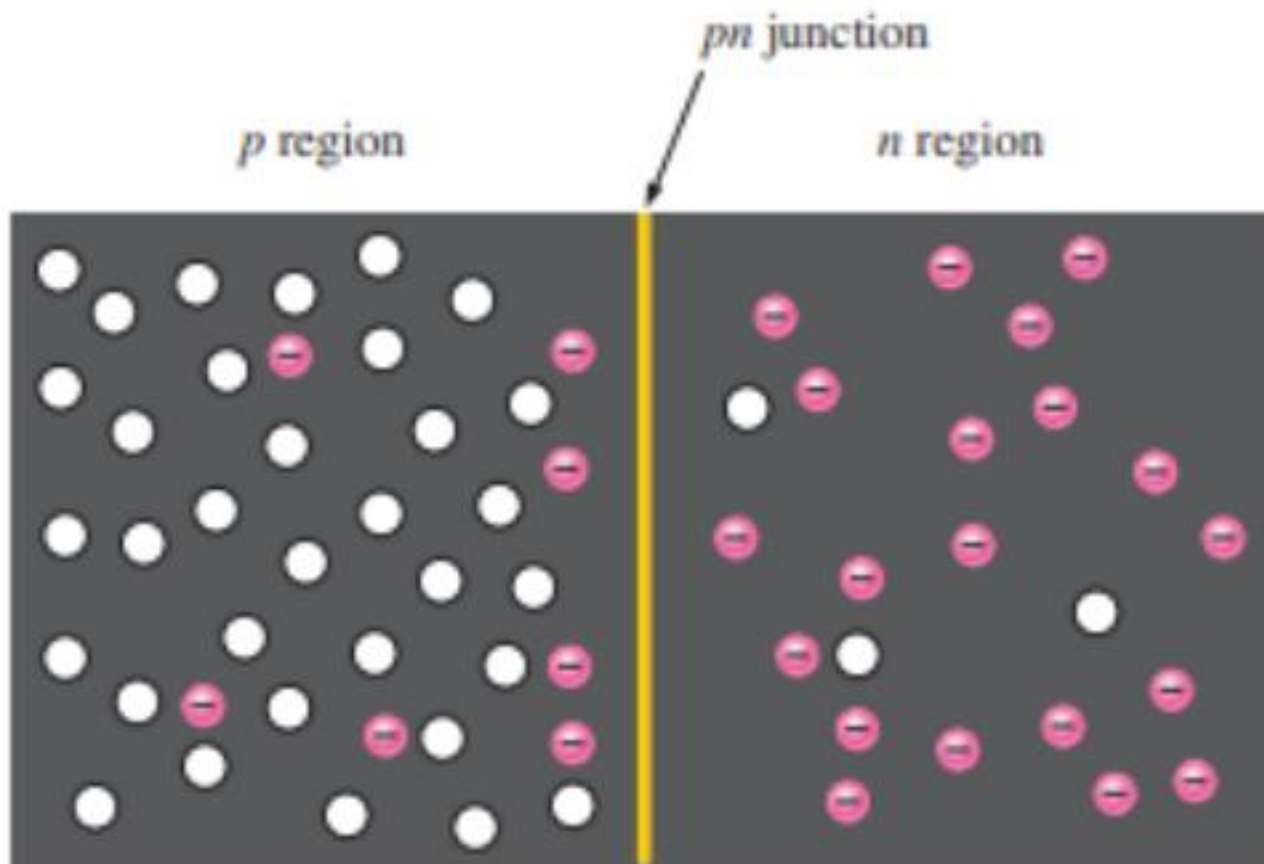
- ▶ Doping occurs when Si or Ge is doped with an element of group III,
- ▶ Produces P-type semiconductors,
- ▶ Also called as acceptor doping.



PN JUNCTION :

- ▶ If n-type and p-type semiconductors are brought together, a pn junction forms at the boundary between the two regions and a diode is created.
- ▶ The ***p region*** has many holes (majority carriers) from the impurity atoms and only a few thermally generated free electrons (minority carriers).
- ▶ The ***n region*** has many free electrons (majority carriers) from the impurity atoms and only a few thermally generated holes (minority carriers)
- ▶ pn junction is the basis for diodes, certain transistors, solar cells, and other devices





- (a) The basic silicon structure at the instant of junction formation showing only the majority and minority carriers. Free electrons in the n region near the pn junction begin to diffuse across the junction and fall into holes near the junction in the p region.

Depletion region:

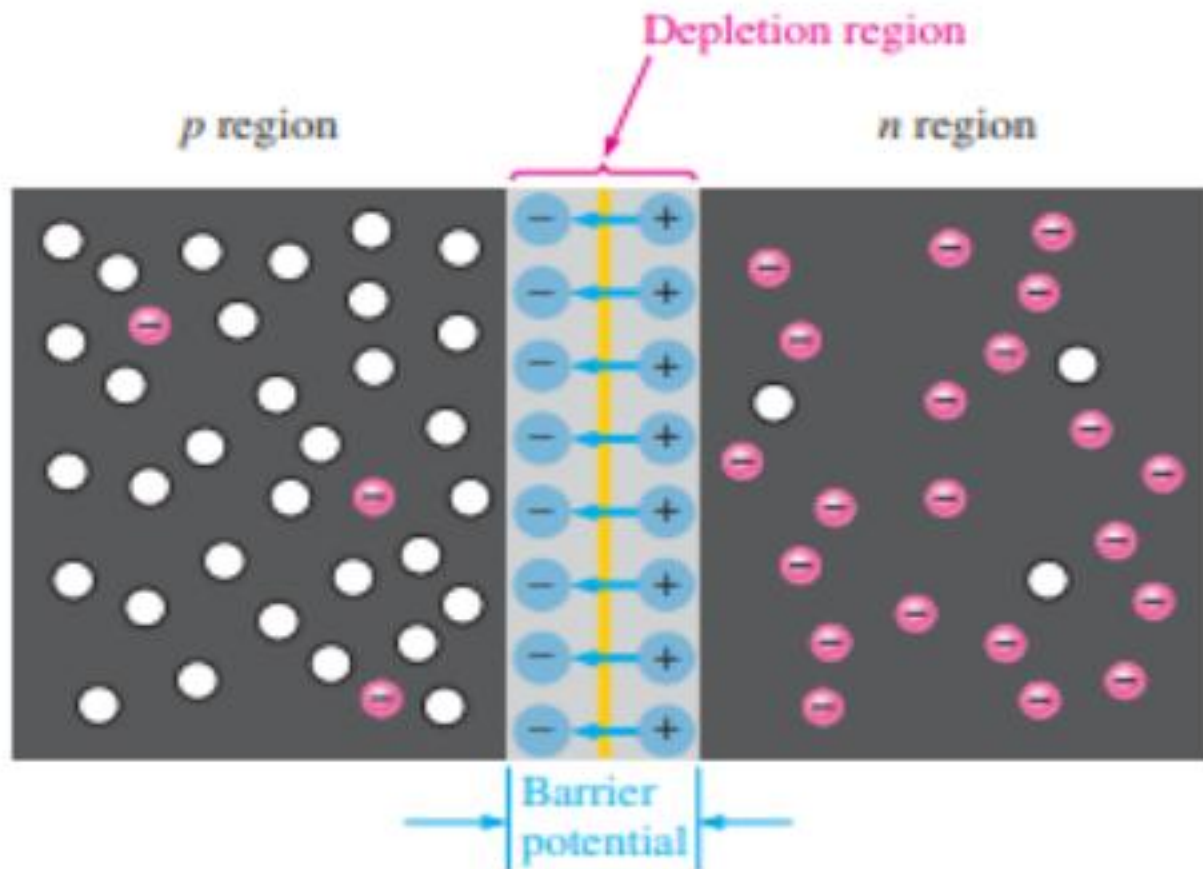
- ▶ In n-type semiconductors, large number of free electrons is present due to this they get repelled from each other and try to move from a high concentration region (n-side) to a low concentration region (p-side).
 - ▶ Hence, the free electrons from n-side attract towards the holes at p-side.
 - ▶ Each free electron that is crossing the junction from n-side to fill the hole in p-side atom creates a negative ion at p-side. Similarly, each free electron that left the parent atom at n-side to fill the hole in p-side atom creates a positive ion at n-side.
 - ▶ The net negative charge at p-side of the p-n junction prevents further flow of free electrons crossing from n-side to p-side because the negative charge present at the p-side of the p-n junction repels the free electrons. Similarly, the net positive charge at n-side of the p-n junction prevents further flow of holes from p-side to n-side.
 - ▶ The depletion region acts like a wall between p-type and n-type semiconductor and prevents further flow of free electrons and holes.
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Barrier Potential:

- ▶ The potential difference of the electric field across the depletion region is the amount of voltage required to move electrons through the electric field. This potential difference is called the ***barrier potential*** and is expressed in volts.
- ▶ The barrier potential is typically 0.7 V for a silicon diode and 0.3 V for germanium.



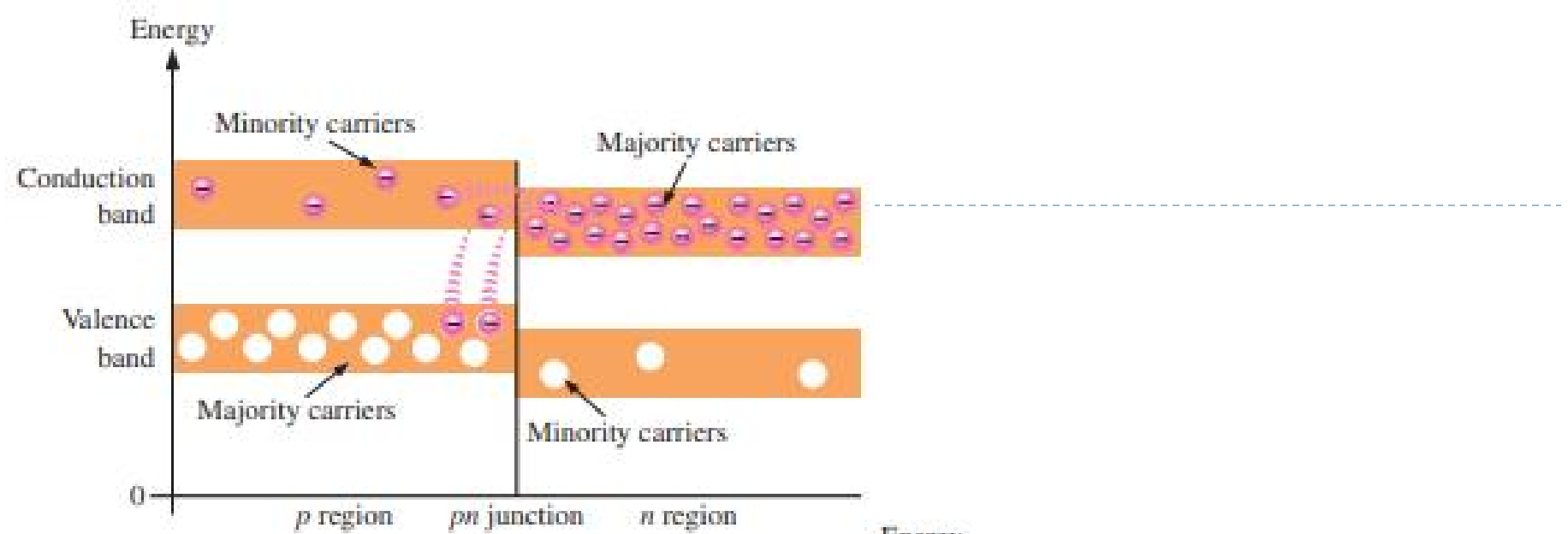


(b) For every electron that diffuses across the junction and combines with a hole, a positive charge is left in the *n* region and a negative charge is created in the *p* region, forming a barrier potential. This action continues until the voltage of the barrier repels further diffusion. The blue arrows between the positive and negative charges in the depletion region represent the electric field.

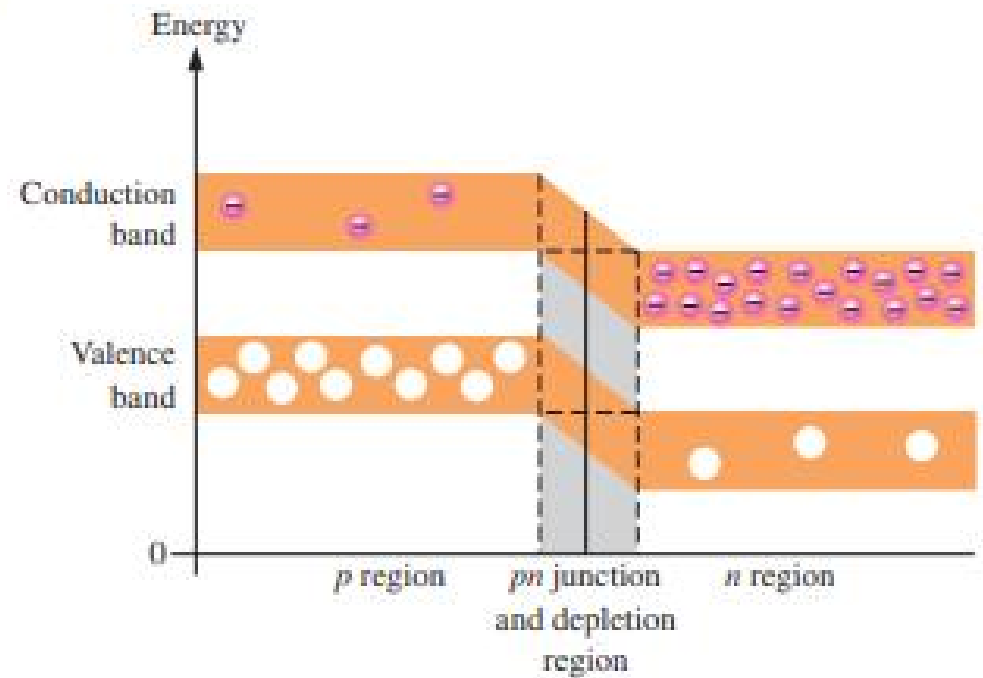
HISTORY NOTE

After the invention of the light bulb, Edison continued to experiment and in 1883 found that he could detect electrons flowing through the vacuum from the lighted filament to a metal plate mounted inside the bulb. This discovery became known as the *Edison effect*.

An English physicist, John Fleming, took up where Edison left off and found that the Edison effect could also be used to detect radio waves and convert them to electrical signals. He went on to develop a two-element vacuum tube called the *Fleming valve*, later known as the *diode*. Modern *pn* junction devices are an outgrowth of this.



(a) At the instant of junction formation



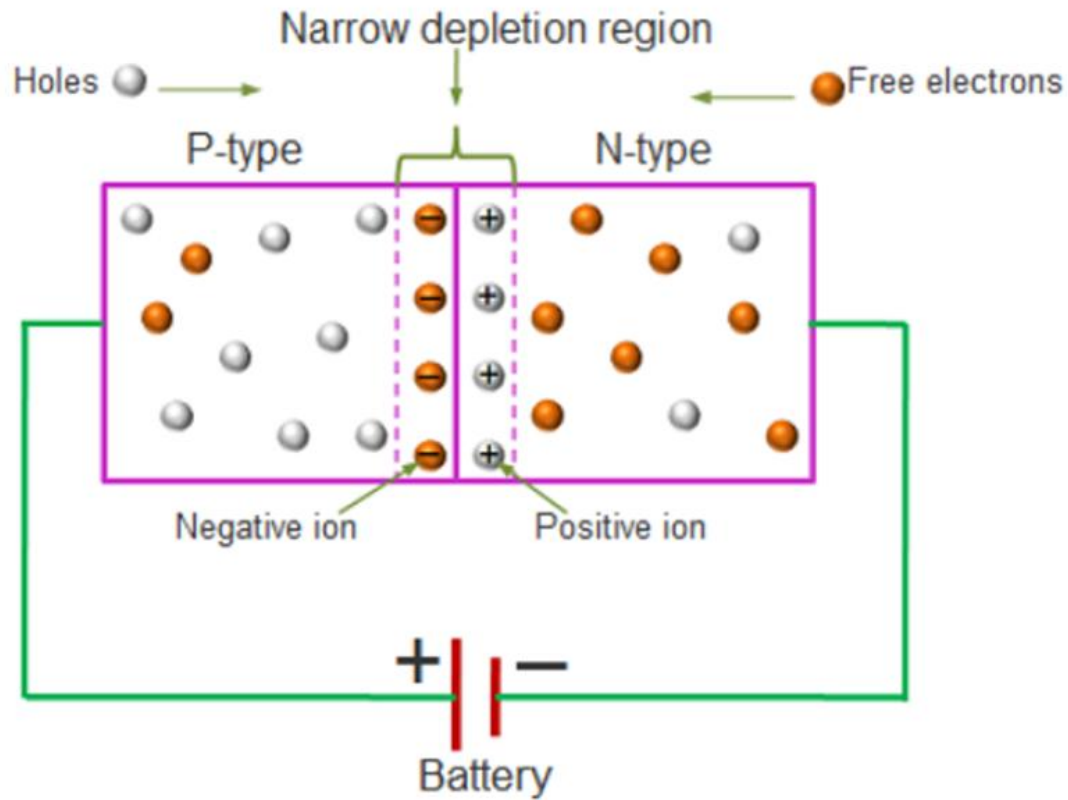
(b) At equilibrium

BIASING:

- ▶ There is current through a diode only when it is forward-biased. Ideally, there is no current when there is no bias nor when there is reverse bias. Actually, there is a very small current in reverse bias due to the thermally generated minority carriers, but this can usually be neglected.
 - ▶ Avalanche occurs in a reverse-biased diode if the bias voltage equals or exceeds the breakdown voltage.
 - ▶ A diode conducts current when forward-biased and blocks current when reversed-biased.
 - ▶ Reverse breakdown voltage for a diode is typically greater than 50 V.
 - ▶ The V-I characteristic curve shows the diode current as a function of voltage across the diode.
 - ▶ The resistance of a forward-biased diode is called the dynamic or ac resistance.
 - ▶ Reverse current increases rapidly at the reverse breakdown voltage.
 - ▶ Reverse breakdown should be avoided in most diodes.
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FORWARD BIASING:

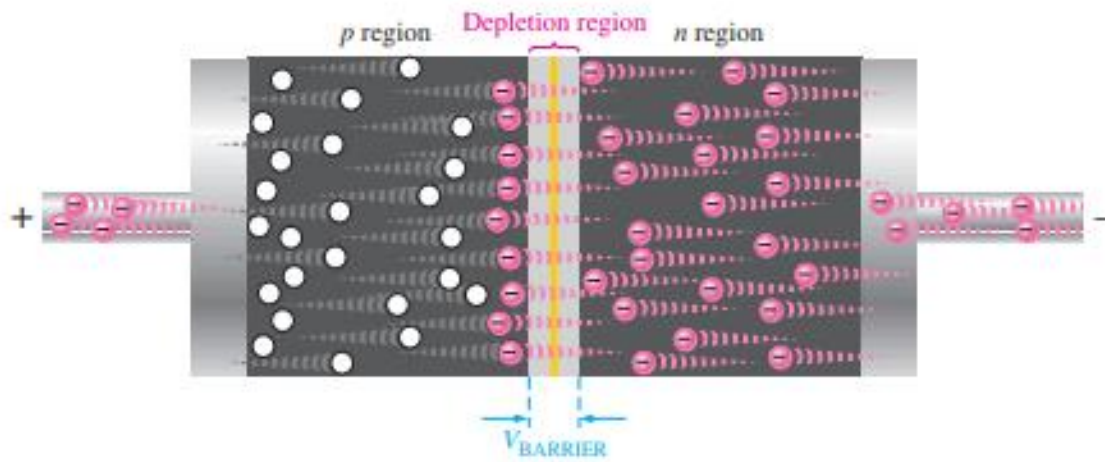


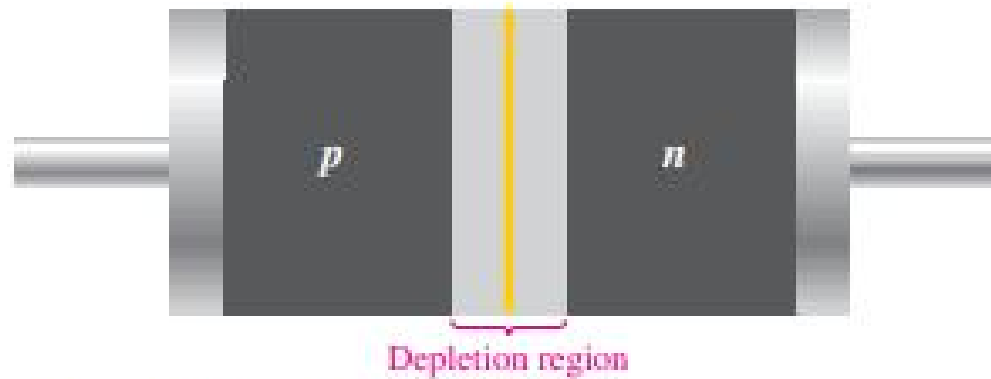
Forward bias



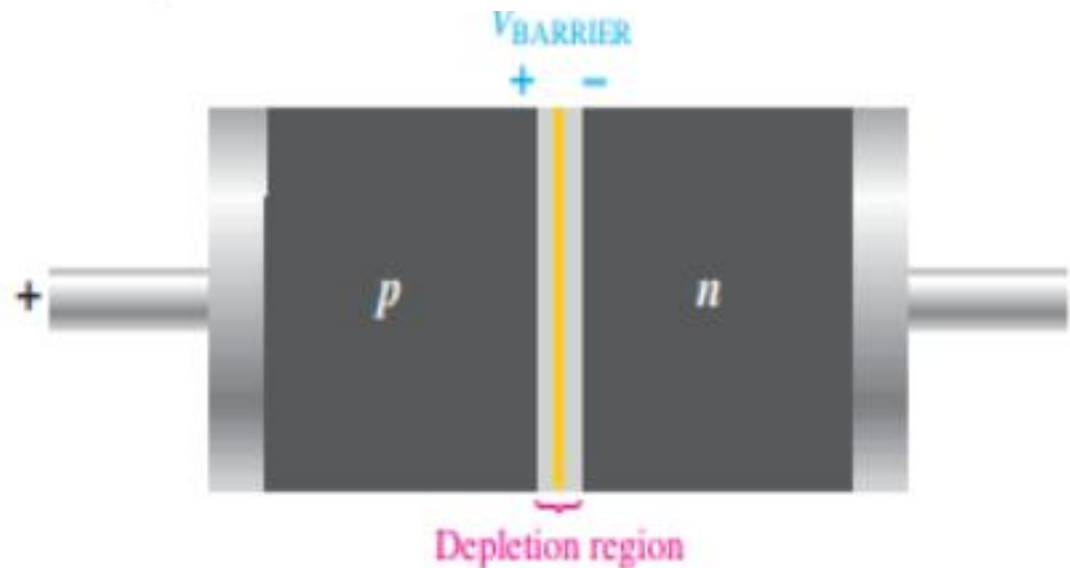
FORWARD BIASING:

- ▶ The condition that's allows current through pn junction,
- ▶ External voltage decreases width of potential barrier.
- ▶ Requirements:
 - ▶ n-region connected to –ve terminal of the battery and p-region connected to +ve terminal of the battery.
 - ▶ Bias voltage must be greater than the barrier potential



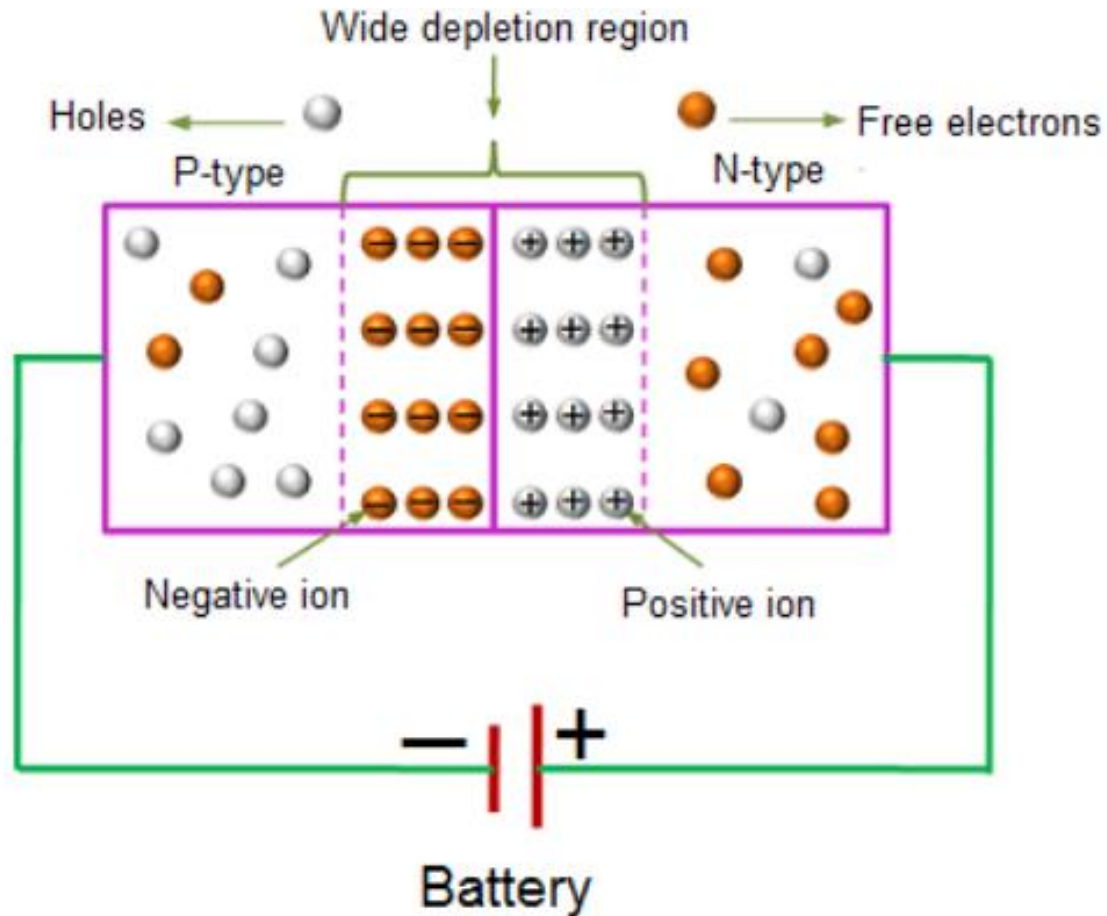


(a) At equilibrium (no bias)



(b) Forward bias narrows the depletion region and produces a voltage drop across the pn junction equal to the barrier potential.

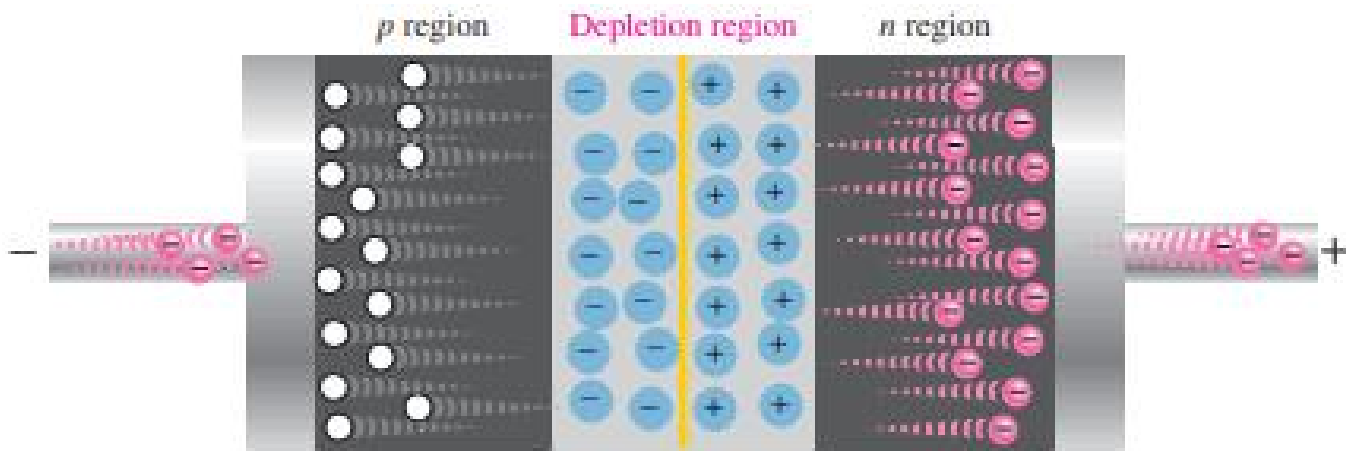
REVERSE BIASING:



Reverse bias

REVERSE BIASING:

- ▶ pn junction blocks current in reverse bias
- ▶ External voltage increases the built-in potential barrier.
- ▶ Depletion region is wider in reverse biasing.
- ▶ Requirements:
 - ▶ p-region connected to +ve terminal of the battery and p-region connected to -ve terminal of the battery.



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- ▶ Reverse current is so small that it can be neglected. However, if the external reverse-bias voltage is increased to a value called the ***breakdown voltage***, the reverse current will drastically increase.
 - ▶ Breakdown voltage is usually greater than 50 volts.



V-I CHARACTERISTIC CURVE:

